Proactive Memory for Event-Driven LLM Agents

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Abstract

Proactive LLM agents anticipate service opportunities from ongoing observations and accumulated history. Existing approaches emphasize recent context or retrieval of related memories, providing limited support for validating recurring behavioral regularities and assessing their applicability to the current event. We propose Proactive Memory (ProactMem), combining bounded recent events, source-checked temporal trajectories, and replay-assessed behavioral regularities. The trajectory validation step checks extracted facts against source events, while historical replay assesses recurring regularities before they are matched to current events. We also construct ProactiveStream, a longitudinal benchmark covering three application scopes, where agents process isolated event streams chronologically under episode-aware opportunity labels. On ProactiveStream, ProactMem achieves an overall effectiveness (F1) of 52.6%, a 21.4% relative improvement over the strongest of five external baselines.

1 Introduction

LLM agents are extending task execution toward proactive assistance that identifies service opportunities from ongoing observations and accumulated history. In activity support, smart homes, and software collaboration, useful assistance depends on recognizing new needs, selecting relevant historical evidence, and tracking prior assistance.

Existing work develops proactive decisions and agent memory. ProactiveAgent learns task proposals from activity contexts (Lu et al., 2025); ContextAgent combines sensed context with historical personas (Yang et al., 2025). Mem0 consolidates and retrieves information (Chhikara et al., 2025), while A-MEM links historical notes (Xu et al., 2025). ProactiveBench and ContextAgentBench evaluate task proposals and contextual service; LoCoMo and LongMemEval assess long-term conversational memory (Maharana et al., 2024; Wu et al., 2025). Continuous assistance also requires assessing memory accumulation, current applicability, and changing opportunities within one event stream. This raises a central question: *how can history become source-grounded, historically supported, and currently applicable memory for timely, selective assistance?*

To study this question, we construct *ProactiveStream*, a longitudinal event-stream benchmark spanning Sports, Home, and Code. Agents build memory chronologically within isolated participant, home, and repository streams, assessing each opportunity's first actionable window and subsequent changes. With a shared backbone and intervention control policy, existing methods show different trade-offs between opportunity recovery and false interventions (Table 2). Figure 1 links earlier evening walks to current assistance.

We propose Proactive Memory (ProactMem; Figure 2), linking recent observations, temporal activities, and recurring behavior. Event memory preserves recent evidence; temporal trajectory memory abstracts source-grounded activities; behavioral regularity memory captures supported conditional behavior. Together, these layers connect immediate changes with long-term context.

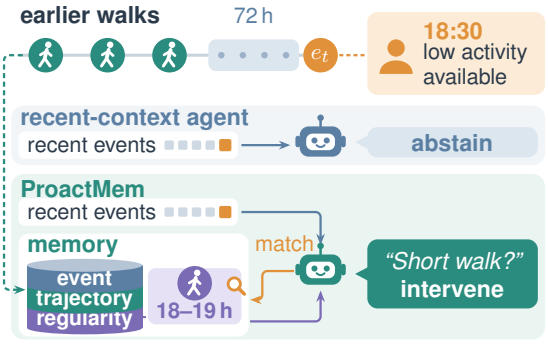


Figure 1: Illustrative Sports scenario. ProactMem retrieves a supported evening-walk regularity to guide assistance beyond recent events.

LLMs construct cited trajectories from completed events, followed by source validation of typed facts. Historical replay assesses regularities inferred from these trajectories. At decision time, contextual assessment combines the current event, recent history, trajectories, and applicable regularities to judge service value. Intervention control filters repeated proposals; the finalized decision precedes event commitment and memory updates.

Our contributions are:

- ProactiveStream evaluates longitudinal memory formation and service decisions across three scopes, preserving causal history, entity isolation, and episode-aware labels.
- ProactMem combines source-checked trajectory abstraction, replay-based regularity assessment, and current-condition matching to supply historical context for proactive service decisions.
- ProactMem uses long-term behavioral context to identify opportunities missed by recent events alone and support more selective interventions.

2 Problem formulation

Each participant, household, or repository u has an isolated public event stream

$$E^u = (e_1^u, \ldots, e_{T_u}^u), \qquad e_t^u = (\tau_t^u, x_t^u), \quad (1)$$

where T_u is the stream length and x_t^u contains public fields available at time τ_t^u. Events follow chronological order with deterministic tie-breaking; we omit u when clear. State Z_t^u comprises prior same-entity observations, finalized outputs, and the memory and intervention-control states derived from them. The policy produces

$$(\hat{y}_t, \hat{m}_t) = \pi(e_t^u, Z_t^u), \qquad \hat{y}_t \in \{0, 1\}, \quad (2)$$

where $\hat{y}_t = 1$ denotes *intervene*, $\hat{y}_t = 0$ denotes *abstain*, and $\hat{m}_t$ is an optional evidence-grounded service message. After finalization, update function U commits the event and output:

$$Z_{t+1}^u = U(Z_t^u, e_t^u, \hat{y}_t, \hat{m}_t). \quad (3)$$

The objective is to detect timely, actionable *service opportunities* supported by current or prior same-entity observations. Evaluation uses fixed reference decisions $y_t \in \{0, 1\}$ under Section 4.2; labels and annotation metadata remain evaluator-private.

3 Proactive Memory

ProactMem uses three complementary memory layers (Figure 2): event memory for recent public events, temporal trajectory memory for source-checked activity summaries, and behavioral regularity memory for replay-assessed conditional regularities. Construction checks source fidelity and historical support. The decision stage combines current observations with applicable historical evidence through contextual assessment and intervention control.

3.1 Memory construction and update

Service opportunities can depend on both recent changes and recurring behavior, motivating a memory hierarchy that preserves fine-grained evidence while validating longer-term abstractions.

Bounded event memory. Evidence utilization can degrade in long contexts (Liu et al., 2024), motivating a bounded event layer that preserves recent observations at their original granularity. For event e_t, event memory retains prior events in their original public representation, intersecting a temporal horizon Δ with capacity K:

$$S_t = \mathrm{Tail}_{K-1}\left((e_i)_{i<t,\, 0 \le \tau_t - \tau_i \le \Delta}\right). \quad (4)$$

Here Tail_{K-1} selects the most recent $K-1$ events in the filtered sequence. We use $\Delta = 72$ hours and $K = 100$: at most 99 prior events accompany the separately supplied current event. A separate buffer retains all committed events from the open local day for trajectory construction.

Fact-validated temporal trajectory memory. To retain activity context beyond the recent window while keeping abstractions traceable, we construct temporal trajectories with source-level fact validation. After a local day closes, a construction LLM summarizes its buffered events as trajectory segments with intervals, typed facts, and event citations. Let $F(\cdot)$ denote normalized checkable facts obtained from typed fields. For segment d and its cited events $\mathcal{E}(d)$, source consistency requires

$$\begin{aligned} F(d) &\subseteq \bigcup_{e \in \mathcal{E}(d)} F(e), \\ \mathcal{E}(d) &\subseteq E^u_{\text{completed day}}. \end{aligned} \quad (5)$$

Here $E^u_{\text{completed day}}$ contains the same entity's events from that completed day. The validator checks entity, interval, citations, and canonicalized scalar

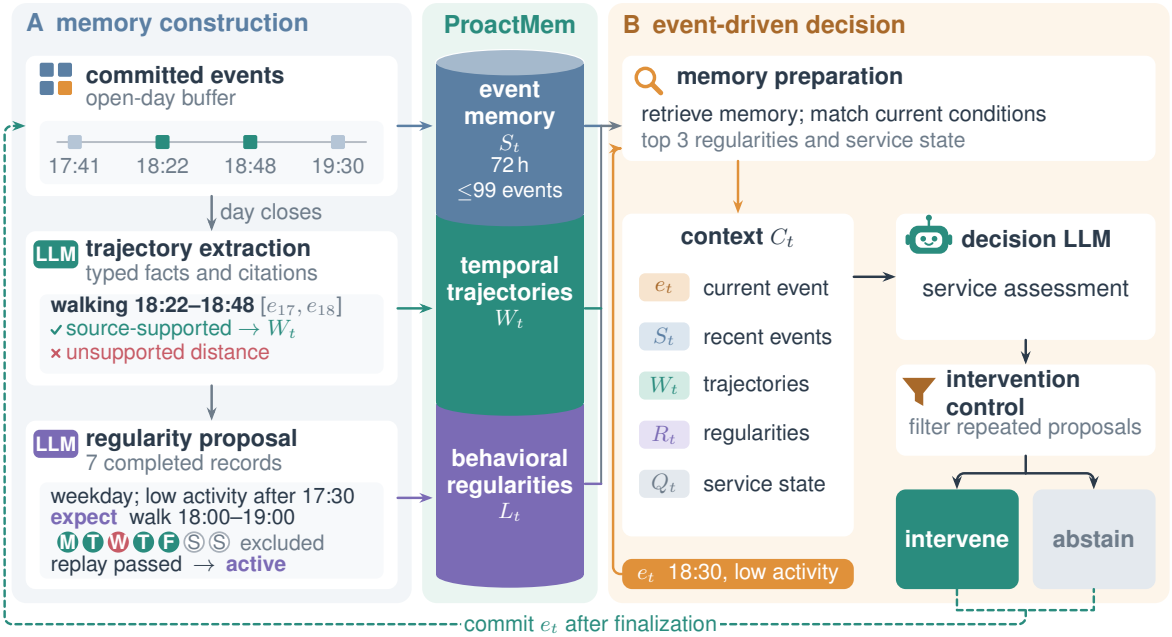


Figure 2: ProactMem architecture. (A) Source validation and replay build hierarchical memory. (B) Retrieved memory and current conditions inform contextual assessment and intervention control. The event is committed after the final decision (dashed arrows).

values at compatible field paths. Acceptance requires source-supported typed facts and resolved citations. Accepted temporal trajectories form W_t, retaining historical context beyond event memory. Appendix B details field handling and buffering.

Replay-assessed behavioral regularity memory. To ground behavioral expectations in observed recurrence, we assess proposed conditional regularities through historical replay and revise their eligibility as evidence accumulates. Weekly consolidation groups seven completed daily records per entity in time order, with each record assigned to one cycle. A cycle may span more than seven calendar days. An LLM proposes a trigger g, expected behavior p, time window, optional weekday restrictions, and source daily identifiers. Replay checks these conditions against validated daily facts: an observable triggered day is fulfilled when p occurs at or after the trigger within the window and violated otherwise. Missing observations are censored, inapplicable days are excluded, and each date contributes at most once. Let n_f count fulfilled (supporting) days and n_v count violated days:

$$\text{conf}_{\text{replay}} = \frac{n_f}{n_f + n_v}, \qquad \text{lift} = \frac{\text{conf}_{\text{replay}}}{\text{base}(p)}. \tag{6}$$

The base rate is the frequency of p over observable days under the same time-window and weekday restrictions across both triggered and other observable days. A zero denominator gives a zero score.

A well-formed proposal becomes *active* when $n_f \geq 3$, confidence ≥ 0.5, and lift ≥ 1.0; otherwise it remains a decision-invisible *candidate*. Malformed proposals are rejected. Consolidation resets the online trajectory view, retaining records for audit. Later observations update regularity confidence and lifecycle: repeated violations can lower retrieval priority or suspend eligibility. Appendix B.2 specifies these transitions.

3.2 Memory retrieval and selection

To make accumulated regularities relevant to the present service opportunity, retrieval checks their trigger conditions, temporal applicability, and occurrence state. The event memory supplies the bounded sequence and W_t supplies all completed temporal trajectories in the current cycle. Retrieval of regularities additionally checks entity-local trigger context, weekdays, and temporal applicability with a 30-minute lead and grace interval. Matching considers current and previously recorded causal trigger occurrences. An occurrence is fulfilled when its expected behavior has been observed; censored occurrences have insufficient observations for assessment. Selection excludes both states. For

regularity store L_t^u, lifecycle eligibility $\mathrm{Eligible}_t$, trigger and time matching Match_t, and occurrence state s_t,

$$R_t = \mathrm{Top3}\big\{\ell \in L_t^u : \mathrm{Eligible}_t(\ell) \wedge \mathrm{Match}_t(\ell), \; s_t(\ell) \notin \{\text{fulfilled}, \text{censored}\}\big\}. \quad (7)$$

The top-three selection ranks eligible regularities by occurrence state, lifecycle, confidence, and temporal distance (Appendix B.2). Retrieval supplies applicable regularities and occurrence states for contextual assessment.

3.3 Proactive decision

Timely assistance requires combining current needs with historical context (Yang et al., 2025).

Contextual assessment and intervention control. For each event, the decision context combines the current event, recent events, temporal trajectories, selected regularities, and observable episode state:

$$C_t = (e_t, S_t, W_t, R_t, Q_t), \quad (8)$$

where Q_t describes the service opportunity's observable state; the controller separately records previously allowed interventions to track service coverage. The LLM produces $(\tilde{y}_t, \tilde{m}_t) = \pi_\theta(C_t)$, proposing intervention when the current event *or* historical context supports concrete, timely, feasible, low-risk assistance, and abstention otherwise.

Within each scope, all methods apply the same intervention controller policy to their own context and intervention history. The controller compares proposed services with previously allowed interventions and converts duplicates to *abstain*. Scope-specific changes in opportunity state, service identity, contextual regularities, or local day permit renewed assessment; each intervention originates from an LLM proposal. The controller finalizes the proposal as $(\hat{y}_t, \hat{m}_t)$, the output defined in Section 2. Appendix B.3 specifies matching fields, phase transitions, and re-entry rules.

Inference stages e_t, retrieves prior memory and matches current conditions, assembles C_t for assessment, applies intervention control, and commits e_t after finalization. Each complete public event enters the event buffer and open-day buffer, advancing trajectory construction, regularity replay, and lifecycle updates. Maintenance incorporating e_t can affect only later decisions. Reference labels remain private during construction and retrieval.

4 ProactiveStream benchmark

ProactiveStream evaluates service-opportunity detection in longitudinal streams as agents accumulate history. Table 1 gives its composition.

4.1 Scopes and public events

Sports uses participant activity and availability contexts for optional walking and everyday physical-activity assistance.

Home uses time-bounded home-sensor observations for gentle, low-risk check-ins. Each sensor segment becomes available at interval closure.

Code uses repository activity at GitHub's native event granularity for low-risk triage, review, and maintenance, with one decision per event.

Each participant, home, or repository defines an isolated chronological stream, allowing earlier observations to inform later service opportunities. Public inputs contain contemporaneously available observations, using the same scope-specific schema for both label classes. Agents construct their own memory from these events; annotation and scoring metadata remain evaluator-private. Appendix A details source datasets, fields, and preprocessing.

4.2 Reference labels and causal replay

Annotators assess e_t and its complete prior same-stream history $\mathcal{H}_{<t}$. Let A indicate a concrete, timely, feasible, low-risk opportunity supported by e_t, B an opportunity at e_t supported by historical observations, and N_t^{ref} reference-episode novelty shared across methods:

$$y_t = 1 \iff \big(A(e_t) = 1 \vee B(e_t, \mathcal{H}_{<t}) = 1\big) \wedge N_t^{\mathrm{ref}} = 1. \quad (9)$$

The reference protocol labels the first reliable actionable window *intervene* and covered windows *abstain*, allowing renewed opportunities when support strengthens, the service changes, or an episode reopens. Event-level labels assess timely detection of these service windows.

LLM-assisted triage prioritizes candidate opportunities; three team members then independently label every event while blinded to evaluated-system outputs, followed by cross-review and adjudication. Appendix D details a separate-team diagnostic of reference sensitivity and episode behavior. Main comparisons fix labels, prompts, memory bounds, and validation settings before evaluation.

scope	primary source	entity	streams	input events	temporal coverage	reference interventions
Sports	HeartSteps	participant	9	1,456	367 observed participant-days	365
Home	CASAS	home	4	1,624	14 consecutive days/home	190
Code	GH Archive	repository	4	1,459	longitudinal repository histories	174
total	three scopes	isolated stream	17	4,539	three longitudinal settings	729

Table 1: Composition and provenance of ProactiveStream. Reference interventions mark service opportunities.

Each stream is replayed chronologically from empty method state, with access to the current event and strictly prior same-stream context. Each decision is finalized before the current event is committed to memory; reference labels are joined after inference for scoring. Scoring includes the initial events and continues as history accumulates.

5 Experimental evaluation

We examine overall effectiveness (**RQ1**), model choice (**RQ2**), and memory organization and validation (**RQ3**), with supplementary checks of reference sensitivity and episode behavior.

5.1 Experimental settings

Baselines. We evaluate five external methods:

- **ProactiveAgent** uses its released prompt and proposal-or-null output contract, with causal history of events and decisions capped at 1,000 events (Lu et al., 2025; THUNLP, 2026).
- **ContextAgent** adapts proactive-service judgment to public events (Yang et al., 2025).
- **Mem0** consolidates and retrieves same-entity memories (Chhikara et al., 2025; Mem0, 2026).
- **A-MEM** updates linked notes and associations for relevant-note retrieval (Xu et al., 2025).
- **Graphiti** stores temporal event relations and retrieves same-stream facts (Zep, 2026).

Metrics. Each event receives one post-controller decision; immediate and scheduled services count as *intervene* at the decision event. We report **relevance** (precision), the share of interventions matching reference opportunities; **coverage** (recall), the share of opportunities recovered; and **effectiveness** (F1), their harmonic mean. Overall effectiveness pools confusion counts across all streams and scopes. The **intrusion** score (false positive rate) measures the share of reference negatives receiving interventions; **balance** (balanced accuracy) averages coverage and the correct-abstention rate on reference negatives. Higher relevance, coverage, effectiveness, and balance and lower intrusion are preferred. The **frequency** metric measures interventions per reference-positive event: $(\mathrm{TP}+\mathrm{FP})/(\mathrm{TP}+\mathrm{FN})$, with counts pooled across streams within each scope. Appendix B.4 gives metric formulas and frequency interpretation.

Implementation details. The primary model is `gpt-5.6-luna`, with temperature zero and thinking disabled; ProactMem uses the official OpenAI API. Methods share task semantics and the intervention controller policy, with interface-specific prompts and budgets. The event-only, event and trajectory memory, and ProactMem variants share their decision prompt, schema, and context bound of 72 hours and at most 100 events, including the current event. A-MEM and Graphiti augment that bound with their representations. For ProactMem, each event undergoes contextual assessment with prepared memory and current-condition matches, followed by intervention control. Appendix B.4 lists detailed configurations.

5.2 Main results (RQ1)

ProactiveStream evaluates service decisions as history accumulates.

Comparisons with external methods. ProactMem achieves higher effectiveness and balance across scopes, jointly reflecting opportunity recovery and intervention selectivity (Table 2, RQ1). Appendix D examines reference sensitivity and episode behavior on a separate 450-target sample.

5.3 Model comparison (RQ2)

Within-series model scaling. Construction and decision LLMs change together, with all other settings fixed (Table 3). Effectiveness increases and intrusion decreases from GPT-5.6 Luna through GPT-5.6 Terra to GPT-5.6 Sol across scopes. ProactMem engages these backbones in extracting task-relevant details, inducing conditional regularities,

scope	method	frequency (×)	relevance ↑	coverage ↑	effectiveness ↑	intrusion ↓	balance ↑
Sports	ProactiveAgent	2.75	30.0	82.5	43.9	64.5	59.0
	ContextAgent	2.16	31.0	67.1	42.4	50.0	58.6
	Mem0	3.40	28.8	**97.8**	44.5	81.0	58.4
	A-MEM	2.41	30.6	74.0	43.3	56.0	59.0
	Graphiti	2.20	29.3	64.4	40.3	52.0	56.2
	ProactMem	2.15	**35.8**	76.7	**48.8**	**46.1**	**65.3**
Home	ProactiveAgent	0.78	34.2	26.8	30.1	**6.8**	60.0
	ContextAgent	1.38	35.9	49.5	41.6	11.7	68.9
	Mem0	0.73	23.9	17.4	20.1	7.3	55.0
	A-MEM	1.63	34.6	56.3	42.9	14.1	71.1
	Graphiti	1.53	33.4	51.1	40.4	13.5	68.8
	ProactMem	1.52	**48.6**	**73.7**	**58.6**	10.3	**81.7**
Code	ProactiveAgent	4.45	17.5	78.2	28.7	49.7	64.2
	ContextAgent	2.68	22.5	60.3	32.8	28.2	66.1
	Mem0	0.65	39.8	25.9	31.4	**5.3**	60.3
	A-MEM	2.55	30.4	77.6	43.7	24.0	76.8
	Graphiti	2.36	26.8	63.2	37.7	23.3	69.9
	ProactMem	2.24	**40.1**	**89.7**	**55.4**	18.1	**85.8**

Table 2: Full-scope comparison (GPT-5.6 Luna). Frequency is in multiples (×); other metrics are percentages. Bold and underline mark the best and second-best scores per scope, including ties.

model	Sports		Home		Code	
	effectiveness ↑	intrusion ↓	effectiveness ↑	intrusion ↓	effectiveness ↑	intrusion ↓
GPT-5.6 Luna	48.8	46.1	58.6	10.3	55.4	18.1
GPT-5.6 Terra	54.1	39.4	64.4	9.1	60.3	14.9
GPT-5.6 Sol	**58.6**	**34.4**	**69.3**	**7.5**	**66.4**	**11.8**

Table 3: Within-series model scaling (%). Bold and underline mark the best and second-best scores per column.

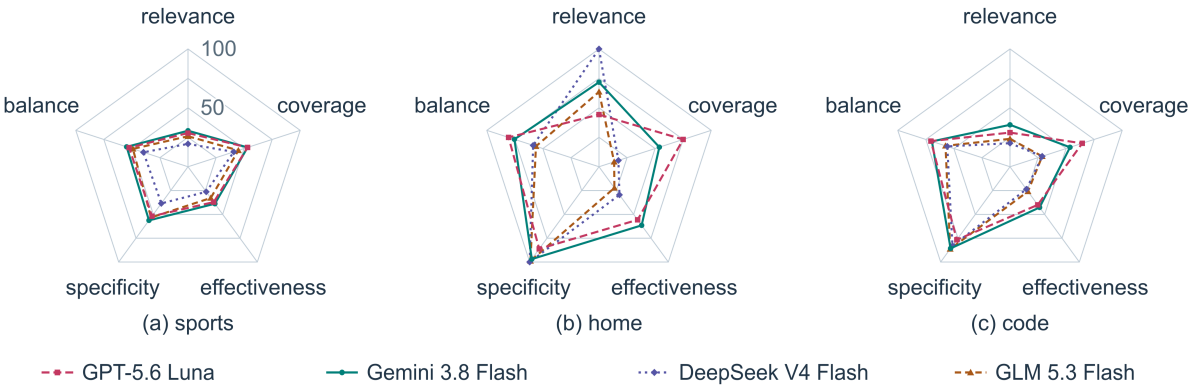


Figure 3: Event-only comparison on selected streams (0–100%). Specificity $= 1 -$ intrusion.

and interpreting historical memory alongside current evidence. The trend is consistent with stronger backbones benefiting this coupled construction-and-decision process. Controlled ablations retain GPT-5.6 Luna.

Cross-family event-only comparison. Using the shared event-only interface, we compare GPT-5.6 Luna, Gemini 3.8 Flash, DeepSeek V4 Flash, and GLM 5.3 Flash on two selected complete streams per scope (Figure 3; Appendix B.5). The results reveal scope-dependent trade-offs between opportunity recovery and correct abstention. On Home and Code, Luna recovers more opportunities than DeepSeek and GLM, while the latter achieve higher specificity. The profiles shift on Sports, linking backbone choice to the service domain and the relative importance of missed opportunities and unnecessary interventions.

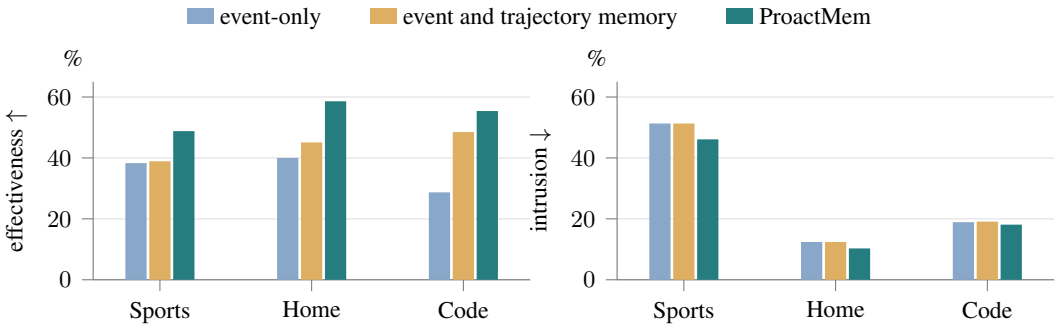


Figure 4: Proactive agent ablation (GPT-5.6 Luna). Effectiveness and intrusion across three memory configurations with a shared decision prompt and intervention control policy.

	Sports		Home		Code	
method	relevance	effectiveness	relevance	effectiveness	relevance	effectiveness
w/o replay gate	31.1	42.7	43.6	54.4	33.8	48.1
w/o trajectory validation	<u>34.4</u>	<u>47.5</u>	<u>45.9</u>	<u>56.7</u>	36.1	51.5
before suppression	–	–	–	–	<u>36.7</u>	<u>53.6</u>
ProactMem	**35.8**	**48.8**	**48.6**	**58.6**	**40.1**	**55.4**

Table 4: Validation and Code intervention control (GPT-5.6 Luna, %). Before suppression scores recorded proposals with later states fixed. Bold and underline mark the best and second-best scores.

5.4 Ablation and memory analysis (RQ3)

Proactive agent ablation. Figure 4 compares event-only memory, event and trajectory memory, and the full hierarchy using Luna under shared decision settings, with representation-dependent intervention keys. Trajectories extend activity context beyond recent events, while regularities organize recurring evidence around its conditions of applicability. Opportunity recovery increases with trajectory memory, especially on Code; the full hierarchy combines further recovery with fewer false interventions across scopes. This progression is consistent with the complementary roles of activity-level context and conditional behavioral evidence in the complete system (RQ3).

Memory validation. Source validation checks trajectory fidelity; replay assesses historical support for conditional regularities. Table 4 compares both ablations with the same model and decision settings. *w/o replay gate* retains current matching and bypasses activation thresholds for well-formed, source-linked candidates reconstructed online. *w/o trajectory validation* retains schema-valid same-entity extractions, weekly construction, and replay, omitting fact checks. Both reduce relevance and effectiveness; removing replay yields the larger effectiveness decline. On Code, trajectory validation is associated with fewer false interventions at unchanged coverage (RQ3).

Intervention control. The Code analysis reveals a trade-off between coverage and selectivity. On recorded proposals, suppression removes more false positives than true positives, improving relevance and effectiveness while reducing coverage. Subsequent states and decisions remain fixed (Appendix C).

6 Related work

Agent memory. Generative Agents, MemoryBank, and MemGPT organize experience through reflection, updating, or tiered context management (Park et al., 2023; Zhong et al., 2024; Packer et al., 2023). Mem0, A-MEM, and HippoRAG use extracted memories, linked records, and graph retrieval (Chhikara et al., 2025; Xu et al., 2025; Gutiérrez et al., 2024). Graphiti preserves temporal validity and provenance in event-derived relations (Zep, 2026).

Proactive agents and memory. ProactiveAgent studies task proposals and silence using annotated events (Lu et al., 2025). ContextAgent combines sensory contexts with historical personas to anticipate service needs (Yang et al., 2025), while PASK couples streaming demand detection with hybrid long-term memory (Xie et al., 2026). MemCog (Li et al., 2026) explores memory for dialogue reasoning, while Remember When It Matters injects selected reminders through a separate memory

agent for long-horizon action (Wu et al., 2026). ProactMem grounds trajectory-derived regularities through source checks, replay, and event matching.

Evaluation of proactive service. LoCoMo and LongMemEval assess long-term conversational memory and reasoning (Maharana et al., 2024; Wu et al., 2025). Proactive benchmarks examine task proposals, latent-need detection, or memory triggering (Lu et al., 2025; Xie et al., 2026; Li et al., 2026). ProactiveStream evaluates sequential *intervene* or *abstain* decisions with accumulating memory in isolated, causally replayed streams, using episode-aware references for intervention timing. Interruption cost depends on task phase and receptivity (Adamczyk and Bailey, 2004; Iqbal and Bailey, 2007, 2008; Mehrotra et al., 2016), motivating evaluation of opportunity recovery and intrusion.

7 Conclusion

ProactMem connects recent events with source-grounded trajectories and historically supported regularities for proactive assistance. On ProactiveStream, it achieves higher effectiveness and balance across all three scopes than the evaluated baselines. Together, the framework and benchmark link memory formation with the changing relevance of historical evidence in continuous service decisions.

Limitations

ProactiveStream covers low-risk service settings in Sports, Home, and Code under a protocol fixed after joint method and benchmark development. Independent held-out evaluation and cross-family transfer of constructed memory remain future work. Reference labels represent service opportunities under the annotation protocol; the 450-target review assesses three selected streams. The Code suppression diagnostic holds later states fixed; controller-free behavior requires a sequential comparison. Historical replay assesses observed support; prospective validity under changing behavior requires further study. Our offline metrics evaluate opportunity detection. Recipient benefit, interruption burden, end-to-end cost, and deployment safety require prospective measurements. The reported results support comparisons within the evaluated protocols, with broader generalization depending on additional populations, observation conditions, and model families.

Ethical considerations

Our evaluation is limited to optional activity support, gentle check-ins, and low-risk repository assistance. Medical and high-risk safety decisions require separate domain-specific validation and safeguards. Inference is isolated by stream, and reference labels, episode annotations, and scoring metadata are excluded from model inputs.

Longitudinal events and derived memories can reveal sensitive routines even when identifiers are pseudonymized. Event content, prediction traces, and inferred trajectories require access and retention protections; external-model use should limit data transmission and respect source permissions.

Deployment should require informed opt-in and allow inspection, correction, deletion, retention control, and disabling proactive assistance. An intervention prediction represents a service proposal; consequential action requires separate authorization. These deployment safeguards require separate implementation and evaluation.

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A Benchmark and annotation protocol

A.1 Public events and availability

Sports. HeartSteps (Klasnja et al., 2019) supplies 1,456 mobile activity meta-events from nine complete participant streams over 367 observed participant-days. Public fields include decision-time availability, recognized activity, recent steps, location category, weather, and local decision slot. Services concern optional activity support. Inputs omit future outcomes and randomized treatments.

Home. CASAS (Cook et al., 2013) supplies ambient-sensor traces from four homes over 14 consecutive days each. Per local day, ten-minute source blocks are assigned exactly once to 28 regular 50-minute segments and one 40-minute tail, yielding 406 events per home. A segment becomes available only after interval closure and contains within-interval motion counts, room sequences, door actions, and compact numeric summaries. Gentle check-ins exclude disease, fall, travel, danger, and unsupported preference inference.

Code. Public repository activity, including GH Archive records (Grigorik, 2026), supplies 1,459 events across four repositories with 174 reference interventions. Pushes, issue updates, pull-request actions, and reviews retain GitHub's native event granularity, with one decision per event. Records are ordered by timestamp and then event ID. Each repository owns isolated memory and intervention state even when contributors overlap. Services cover low-risk triage, review, and maintenance; native `PushEvent` records enter the stream.

Public-input contract. Both label classes share the same scope-specific schema: event and stream identifiers, timestamps, and contemporaneously observable fields. Reference labels, candidate flags, episode annotations, memory-level markers, and scoring masks remain evaluator-private. Evaluation includes early events before temporal trajectories or behavioral regularities are available.

A.2 Reference labels and causal replay

Annotators apply the opportunity and novelty semantics in Section 4.2 to each current event and its complete prior same-stream history, independently of a tested method's memory. LLM-assisted triage prioritizes the review queue; three team members independently assign labels while blinded to evaluated-system decisions, followed by cross-review and adjudication of every public event.

The Home aggregation rule precedes event construction. Each evaluation version fixes the event inventory, labels, intervention prompt, event memory bound, and validation settings before main runs on the jointly developed benchmark.

Every stream starts from empty method state and is replayed chronologically under the causal execution order in Section 3.3. Current events are committed only after the output is finalized; maintenance incorporating an event affects later decisions. Labels are joined offline; reference episode states and scoring metadata stay private.

B Implementation and experimental configuration

B.1 Trajectory evidence and replay

The trajectory construction step uses an independent buffer of all committed events from the open local day, cleared after processing. Activity segments contain a local interval, typed `observed`, `context-before`, and `outcome` fields, plus source-event identifiers. The checks in Equation 5 canonicalize scalar values, accept exact or schema-preserving suffix field paths, and require consistent entities, intervals, and resolved citations. Validation and replay operate on typed evidence. Accepted trajectories remain in the current construction cycle after their events leave event memory.

Empty or unobservable records also advance the seven-record construction cycle. Replay follows Equation 6: a triggered, observable day is fulfilled when the expected behavior occurs at or after the trigger within the window; otherwise it is violated. Missing observations are censored, each date counts at most once, and the base rate includes triggered and other observable days under the same time and weekday restrictions. Confidence and lift have zero-denominator scores of zero.

B.2 Lifecycle and current matching

Post-admission confidence combines replay and online counts in a support ratio smoothed with one initial success and failure; censored observations preserve both counts. Two consecutive violations move an active regularity to *weakening*, lowering retrieval priority; four move it to *dormant*. A fulfilled observation restores a weakening regularity to active at confidence ≥ 0.5. Dormant regularities return to candidate for replay readmission.

Current matching. Current matching is deterministic. Retrieval admits upcoming, pending, or deviated occurrences of active or weakening regularities. Occurrence priority is deviated, followed by pending and upcoming.

B.3 Intervention control

In Sports and Home, the intervention key combines scope-specific episode state, proposed service action and target, and sorted regularity identifiers available in context. Plain summaries or graph facts supply an empty regularity-identifier list. Sports state includes availability and activity; Home state includes activity and room and onset, confirmed, and sustained phases at one, two, and three or more consecutive matching observations. Without an explicit episode key, scope-specific time or target fields identify the opportunity, with unchanged evidence additionally required for duplicate suppression. State changes, sequence restarts, changed service or regularity identifiers, and local-day transitions can generate new keys.

Code uses repository-object identity and the most recent allowed intervention's mode, recipient, content, and firing condition. An exact match converts a repeated `create` proposal to *abstain*; changed signatures and other episode operations pass this check. Representation-dependent key evidence can yield different suppression decisions across memory interfaces.

B.4 Model and method interfaces

Decision and memory-construction calls use the model, API, and decoding settings in Section 5.1, with separate output budgets. Decision limits are interface-specific (256 or 512 tokens); construction limits are configured by call role. Table 5 consolidates the external-method interfaces. Each method retains its prompts, retrieval, and budgets.

Graphiti supplies relation facts directly, using the task policy and output contract with memory-specific guidance; its ingestion episodes correspond to source records.

Scoring. Using final-decision confusion counts, relevance is $\frac{\mathrm{TP}}{\mathrm{TP+FP}}$, coverage is $\frac{\mathrm{TP}}{\mathrm{TP+FN}}$, effectiveness is $\frac{2\mathrm{TP}}{2\mathrm{TP+FP+FN}}$, and intrusion is $\frac{\mathrm{FP}}{\mathrm{FP+TN}}$. Balance is $\frac{1}{2}[\frac{\mathrm{TP}}{\mathrm{TP+FN}} + \frac{\mathrm{TN}}{\mathrm{TN+FP}}]$. For frequency, $1\times$ indicates equal counts of interventions and reference-positive events; larger and smaller values indicate more and fewer interventions, respectively. Frequency is interpreted jointly with relevance, coverage, and intrusion.

B.5 Cross-family event-only configuration

Table 6 lists the two complete streams selected within each scope. Streams are ranked by SHA256 of `seed|domain|public_stream_id`. The fixed seed is `proactivestream-event-only-nine-streams-20260925-v1`; the two lowest hashes in each scope determine selection. The selection uses public stream identifiers.

Each stream is replayed chronologically from empty state. Models share the scope-specific task prompt, intervention control policy, and a causal context containing the current event and up to 99 prior same-stream events within 72 hours. Requests use temperature zero and request disabled thinking, with decision output limits of 256 tokens for Sports and Home and 512 for Code. Output-format and transport-recovery settings follow each backend's interface. Final decisions are scored offline against the common reference labels.

C Additional ablation and intervention control results

Table 7 supplements the relevance and effectiveness scores in Table 4 with frequency and coverage. On Code, suppression removes 66 false-positive and 17 true-positive proposals; coverage falls from 99.4% to 89.7%. The diagnostic scores recorded proposals while holding later states fixed. In Figure 4, the event and trajectory memory variant records 4, 15, and 64 more true positives than event-only on Sports, Home, and Code, respectively; ProactMem records a further 57, 33, and 22.

D Reference and episode diagnostics

D.1 Reference sensitivity

The diagnostic selects 150 consecutive targets from one stream per scope using a fixed SHA256 ranking of eligible streams and contiguous starts based only on public metadata, with at least seven days of preceding history. The streams are Sports participant_35, Home home_0004, and Code repository `callstack/react-native-pager-view`. Their target periods are December 16, 2015–January 18, 2016; June 23–28, 2011; and June 21–November 4, 2025. All 37, 203, and 152 preceding events are processed chronologically, respectively: 842 public events supply context for 450 scored targets.

method	interface and settings
ProactiveAgent	Released prompt and proposal/null contract on the study backbone; causal event–decision history capped at 1,000 events per request including the current event (Lu et al., 2025; THUNLP, 2026).
ContextAgent	In-context proactive-service judgment on public events (Yang et al., 2025).
Mem0	Open-source `mem0ai==2.0.12` extraction, consolidation, and retrieval; local `all-MiniLM-L6-v2` embeddings; top-8 same-entity retrieval with search threshold 0.1 (Chhikara et al., 2025; Mem0, 2026).
A-MEM	Linked notes with association updates augment bounded context (Xu et al., 2025).
Graphiti	Graphiti 0.30.2 with Neo4j and local 384-dimensional `all-MiniLM-L6-v2` embeddings; committed events populate stream-isolated graphs. Current-event queries use native hybrid edge search with reciprocal-rank fusion, retrieving up to ten relation facts with validity fields and source IDs (Zep, 2026).

Table 5: External-method memory interfaces and implementation settings.

scope	source stream	events
Sports	HeartSteps participant 05	184
	HeartSteps participant 35	191
Home	CASAS home 0004	406
	CASAS home 0003	406
Code	`Unitech/pm2`	365
	`thephpleague/commonmark`	347
total	six complete streams	1,899

Table 6: Selected complete streams for the cross-family event-only comparison.

scope	method	frequency (×)	coverage ↑
Sports	w/o replay gate	2.19	<u>68.2</u>
	w/o trajectory validation	2.23	**76.7**
	ProactMem	2.15	**76.7**
Home	w/o replay gate	1.65	72.1
	w/o trajectory validation	1.62	**74.2**
	ProactMem	1.52	<u>73.7</u>
Code	w/o replay gate	2.47	83.3
	w/o trajectory validation	2.48	<u>89.7</u>
	before suppression	2.71	**99.4**
	ProactMem	2.24	<u>89.7</u>

Table 7: Additional validation and Code intervention control scores. Frequency uses multiples (×); coverage uses percentages. Bold and underline mark the best and second-best coverage per scope, including ties.

A separate three-reviewer team independently labels each target with its prior same-stream history while blinded to evaluated-system outputs, then discusses judgments to obtain consensus references. Earlier references use original Sports and Home labels and 122 original plus 28 additional Code annotations. Relative to these references, Sports and Home each have 13 changes from *intervene* to *abstain*, with 4 and 0 changes in the reverse direction, respectively; Code has 6 removals and 16 additions to positive labels.

Table 8 scores identical recorded ProactMem outputs against these two sets of references. The predictions come from separate diagnostic runs on the selected targets. The diagnostic uses Luna with temperature zero, thinking disabled, decision limits of 256 tokens for Sports and Home and 512 for Code, and the same context bound of 72 hours and at most 100 events, trajectory validation, replay gating, and intervention control policy. Immediate and scheduled services both count as *intervene*.

Rescoring against the separate-team references yields similar effectiveness on Sports and lower effectiveness on Home and Code relative to the earlier references. Absolute scores are label-sensitive, especially on Home and Code.

D.2 Episode references and matching

A model-assisted post-hoc analysis on these targets reconstructs goal-and-object episodes from positive event anchors. Each reference specifies a service goal and object, causal evidence, onset, explicitly listed valid observations, and any deadline or closure evidence. Semantic matching uses these fields to associate interventions with reference episodes.

Matching and follow-up. Episode coverage follows ProactMem's own interventions, initialized from recorded prefix outputs. Only target interventions are scored. An intervention must match the goal and object at a listed valid observation. Its proposed service time is the decision time plus its specified delay (zero for immediate service), and must meet any explicit episode deadline. The first timely match covers the episode. Each output receives at most one credit: the first uncovered compatible episode in the fixed reference order, otherwise the first already covered one.

Subsequent matches are compared with the method's preceding assistance for that episode. A valuable follow-up requires new causal evidence

scope	positives	I	TP	FP	FN	TN	earlier diagnostic effectiveness	separate-team effectiveness
Sports	27	86	24	62	3	61	42.6	42.5
Home	15	39	15	24	0	111	68.7	55.6
Code	34	42	18	24	16	92	57.6	47.4

Table 8: Reference sensitivity on 150 targets per scope. Effectiveness (%) scores identical outputs under earlier and separate-team references. Positives and confusion counts use the latter; I counts final interventions.

scope	covered/new (coverage)	I	F	V	R	unmatched	late	unresolved	SR	RIR
Sports	23/26 (88.5)	86	23	1	1	60	0	1	27.9	1.2
Home	9/9 (100.0)	39	10	6	17	6	0	0	41.0	43.6
Code	14/31 (45.2)	42	14	3	4	16	4	1	40.5	9.5

Table 9: ProactMem episode diagnostic on 150 targets per scope. Coverage, SR, and RIR use percentages. Unresolved denotes ambiguous semantic or condition matches; $U = 0$ in all scopes.

that changes the actionable service or next step, and a message addressing that change. Rephrasing or continued status alone is redundant; value requiring unobserved delivery, recipient response, or completion remains uncertain. Each reference episode receives coverage credit once.

Metrics and boundaries. Let I count all target interventions, F first valid interventions across all reference episodes, V valuable follow-ups, R redundant follow-ups, and U uncertain follow-ups. New reference episodes have onsets within the target interval. Episode coverage divides covered new episodes by all new reference episodes; supported relevance is $\mathrm{SR} = (F+V)/I$, and redundant intervention rate is $\mathrm{RIR} = R/I$. Uncertain, unmatched, late, and unresolved outcomes contribute only to the intervention count I. Table 9 combines the rates and outcome counts.

Episode coverage is highest on Home and lowest on Code; the redundant-intervention share is highest on Home and lowest on Sports.

The exploratory scores depend on reference construction and semantic matching.